

Slot storage

0xSlots protocol v1 · Slot.sol behind a BeaconProxy · v1 branch, 24 Sept 2026

Each group below, plus OZ Initializable and ReentrancyGuard, has its own ERC-7201 location. **Blue = proposed rename, not in code.**

LOCATION · STRUCT	FIELD	TYPE	HOLDS	WRITTEN BY	PROPOSED
slots.settings Settings → Governance How the slot is governed. Fixed at creation except manager. Renamed so “settings” means only the module’s config.	currency	IERC20	Payment token. Zero address = native ETH.	initialize	
	manager	address	Proposes terms, grants, hands over. Set exactly when something is mutable.	initialize · setManager	
	mutableTax	bool	Tax rate and minimum runway may change.	initialize	
	mutableRecipient	bool	Recipient may change. Also required to add or raise a module fee.	initialize	
	mutableModule	bool	Module may be replaced or removed; new scopes may be granted.	initialize	
slots.terms.tax TaxTerms · live	recipient	address	Receives collected tax, minus the module fee.	initialize · terms land	
	rateBps	uint16	Tax per 30 days, in basis points of the price. 1–10,000.	initialize · terms land	
	minRunwaySeconds	uint32	A buyer’s deposit must cover this long at their price. ≤ 365 days.	initialize · terms land	
slots.terms.module ModuleTerms · live	target	address	The installed module. Zero = none.	initialize · terms land	module
	settings	bytes	abi.encode of the fields the module’s definition lists. Handed to every callback. Empty = none.	initialize · terms land	
slots.module.manifest Manifest → ModuleGrant The slot’s copy. Payouts and callbacks use it; never re-read from the module.	scopes	uint16	Bits: beforeBuy 1 · beforeSelfAssess 2 · afterBuy 4 · afterRelease 8 · afterLiquidate 16 · afterSettle 32 · strict 64 · onInstall 128 · onUninstall 256.	install · scopes land	strict → afterCallbacks-MustSucceed
	feeBps	uint16	Module’s share of collected tax. 0–10,000.	install · grant	
	feeRecipient	address	Where that share is paid.	install · grant	
slots.pending PendingTerms Stored exactly as pendingTerms() returns it. Lands at the next buy or applyTerms, 1 hour after proposal.	taxTerms	TaxTerms	Proposed tax values. Only the fields named in mask count.	proposeTerms · cancel · land	
	moduleTerms	ModuleTerms	Proposed module and its settings, whole.	proposeTerms · cancel · land	
	scopes	uint16	New scopes accepted with grant, waiting for the next buy.	grant · land	
	mask	uint8	Queued terms: TAX_RATE 1 · RECIPIENT 2 · MIN_RUNWAY 4 · MODULE 8 · SCOPES 16.	propose · grant · cancel · land	
	proposedAt	uint64	Start of the delay. Any new proposal restarts it.	proposeTerms · grant	
	reviewedManifest	bytes32	Hash of the manifest reviewed at proposal. Differs when it lands → module dropped.	proposeTerms · cancel · land	reviewedGrantHash
slots.occupancy Occupancy Who holds the slot, at what price, funded by how much.	occupant	address	Current holder. Zero = vacant.	buy · release · liquidate	
	since	uint64	When the current tenure began.	buy	occupiedSince matches the view
	tenureId	uint64	Counts tenures. Scopes operator approvals to one tenure.	buy	
	lastSettled	uint64	Tax is accounted up to this timestamp.	every settle	
	taxCarry	uint64	Remainder below one unit of currency, carried to the next settle.	every settle · buy · vacate	
	price	uint256	Self-assessed price. ≤ 2 ¹²⁸ –1.	buy · selfAssess	
	deposit	uint256	Escrow that tax is drawn from.	buy · topUp · withdraw · settle	
slots.ledger Ledger Tax taken, and what could not be paid either way.	collectedTax	uint256	Tax taken from deposits, not yet paid out.	settle · collect · any payout	
	withdrawableOf	address → uint256	Payouts that could not be pushed. Taken with claim.	failed payout · claim	claimableOf withdraw is the deposit
	debtOf	address → uint256	Tax owed beyond an emptied deposit. Repaid first by topUp or buy.	settle · topUp · buy	
	operatorOf	tenure → addr → bool	Who may reprice for the occupant, for one tenure only.	setOperator	

Terms land at the next buy or applyTerms, 1 hour after proposal. **Settle** runs first in almost every call: tax moves from deposit to collectedTax; a shortfall becomes debt. **Upgrades** append fields only.

Slot methods · writes

Grouped by who may call, then what the slot calls on its module.
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CALLER	METHOD	DOES	PROPOSED
Occupant The current holder. selfAssess also takes an operator.	selfAssess (newPrice)	Restate the price; tax follows from now on. The module may veto.	
	withdraw (amount)	Take deposit back, keeping what the current terms require.	
	release ()	Leave. The rest of the deposit is refunded.	
	setOperator (operator, allowed)	Let someone reprice on your behalf, for this tenure only.	
	applyTerms ()	Land ripe terms without waiting for a buy. Anyone’s call once the slot is vacant.	
Anyone	buy (account, selfAssessedPrice, depositAmount, maxPayment) PAYABLE	Pay the sitting price (nothing if vacant) plus a deposit, and seat account. Lands ripe terms; the module may veto.	
	topUp (amount) PAYABLE	Add to the occupant’s deposit. Repays their debt first.	
	liquidate ()	Evict an occupant whose deposit is empty. No bounty; the reward is a vacant slot.	
	collect ()	Pay out collected tax: the module fee, then the recipient.	
	claim (account)	Send account a payout that could not be pushed to them.	
	multicall (bytes[] data)	Several calls in one transaction, e.g. liquidate then buy.	
Manager Only on a slot created with something mutable.	proposeTerms (taxTerms, moduleTerms, mask)	Queue changes; they wait 1 hour. Checks the module’s settings and pins the manifest it reviewed.	
	cancelTerms (mask)	Drop the named queued changes, leaving the rest.	
	grant (Manifest expected)	Accept what the module declares now. A fee change applies at once; new scopes wait for the next buy.	grantModuleRequest (reviewedRequest)
	setManager (next)	Hand the slot to another manager, immediately.	
Factory Once.	initialize (SlotInit)	Set governance and terms, check the module’s settings, copy its manifest, call onInstall.	
Slot → module ISlotModule. Only callbacks named in the slot’s scopes are called.	checkSettings (bytes settings) VIEW	Revert if the settings are invalid. At proposal and install.	validateSettings
	manifest (bytes settings) VIEW	The scopes and fee the module asks for. May depend on the settings. At proposal, install and grant.	requestedGrant
	beforeBuy · beforeSelfAssess (ctx) VIEW	Veto by reverting. Uncapped gas.	
	afterBuy · afterRelease · afterLiquidate · afterSettle (ctx)	Told after the fact. 500k gas; a revert is ignored unless strict.	ctx.owed/paid → taxOwed/taxPaid
	onInstall (ctx)	Attached. A revert is ignored unless strict.	
	onUninstall (ctx)	Being replaced or removed. Always capped, a revert always ignored.	
	definition () OPTIONAL	JSON for UIs (IDescribedModule). The slot never calls it.	uiMetadata IModuleMetadata

Slot methods · reads

All view. Grouped by what they read.
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READS	METHOD	DOES	PROPOSED
Read · terms	terms · taxTerms · moduleTerms	Live terms: all together, or one group.	
	recipient · taxRateBps · minRunwaySeconds · module	One live field each.	
	pendingTerms · hasRipeTerms	The stored PendingTerms · whether it lands at the next buy.	
	manifest()	The slot's copy of the module's manifest.	moduleGrant
	scopes()	That copy's scopes, unpacked to booleans.	moduleScopes
	grantStatus()	The copy beside what the module declares today: (accepted, declared, feeDiffers, scopesDiffer).	moduleGrantStatus (granted, requested, ...)
	currency · manager · mutableTax · mutableRecipient · mutableModule	Governance, one field each.	
Read · occupancy & money	occupant · price · deposit · occupiedSince · tenureId · lastSettled	One occupancy field each.	
	isVacant · isInsolvent · isOperator (operator)	Nobody seated · deposit can no longer cover what is owed · may act for the current occupant.	
	taxOwed · secondsUntillLiquidation	Tax accrued since the last settle · time until the deposit runs out (max if never).	
	collectedTax · withdrawableOf (account) · debtOf (account)	The ledger, one entry each.	claimableOf
	quoteBuy (account, depositAmount)	Exactly what buy will charge: sitting price + deposit + the account's debt.	
Read · floors & bundles	minDepositForBuy (price) · minDepositToHold (price)	Smallest deposit buy accepts, under the terms that will apply · floor under the terms in force.	
	getSlotInfo · getSlotConstants	Everything above in one struct, including hasRipeTerms · the constants below in one struct (currently missing MAX_MIN_RUNWAY).	
	version()	Implementation version: 1.	

PUBLIC CONSTANTS

MAX_PRICE	2 ¹²⁸ -1	MAX_TAX_BPS	10,000	MAX_MIN_RUNWAY	365 days	TERMS_DELAY	1 hour
BASIS_POINTS	10,000	MONTH	30 days	MODULE_GAS	500,000	PAYOUT_GAS	30,000
					MODULE_CALLBACK_GAS_LIMIT		NATIVE_PAYOUT_GAS_LIMIT
TERM_TAX_RATE	1	TERM_RECIPIENT	2	TERM_MIN_RUNWAY	4	TERM_MODULE · TERM_SCOPES	8 · 16